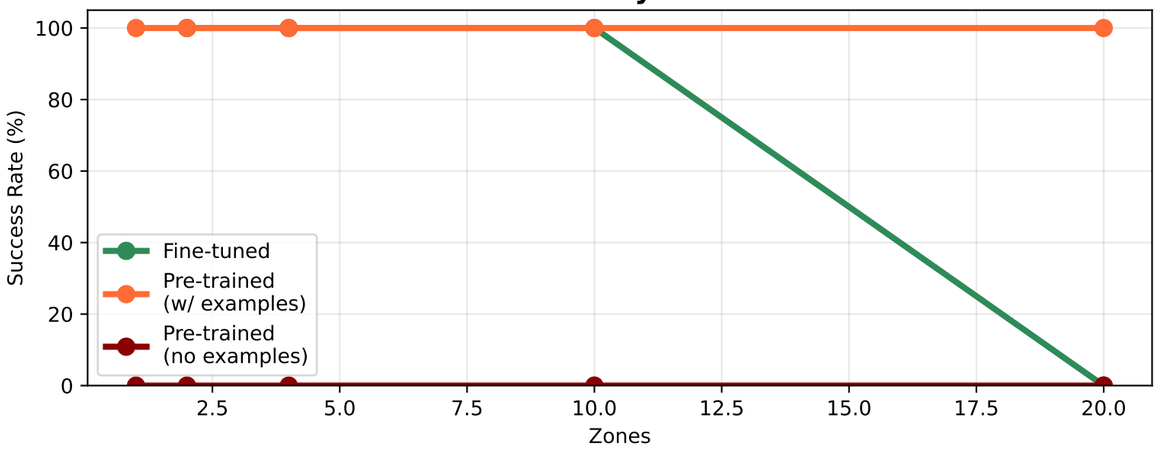
EnergyPlus HVAC Generation: Fine-tuned vs Pre-trained Mistral Model Comparison
Comprehensive Performance Analysis Dashboard

Success Rate by Zone Count



Key Performance Indicators

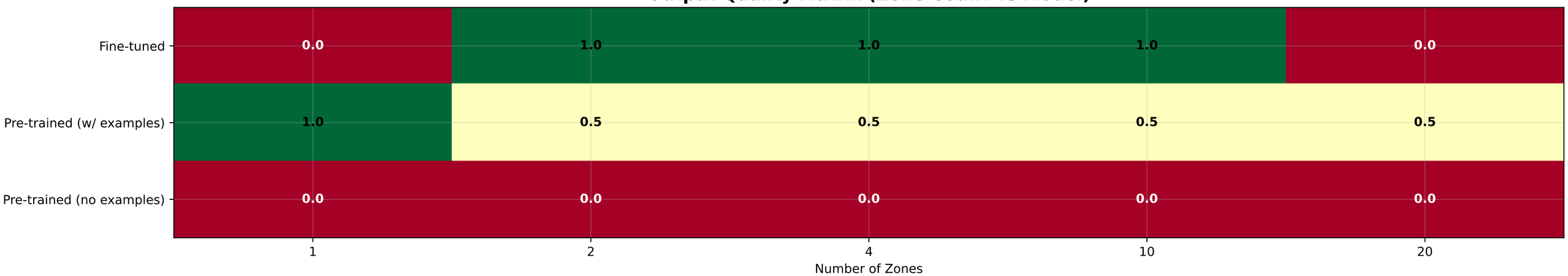
Fine-tuned Success: 75%

Pre-trained Success: 100%

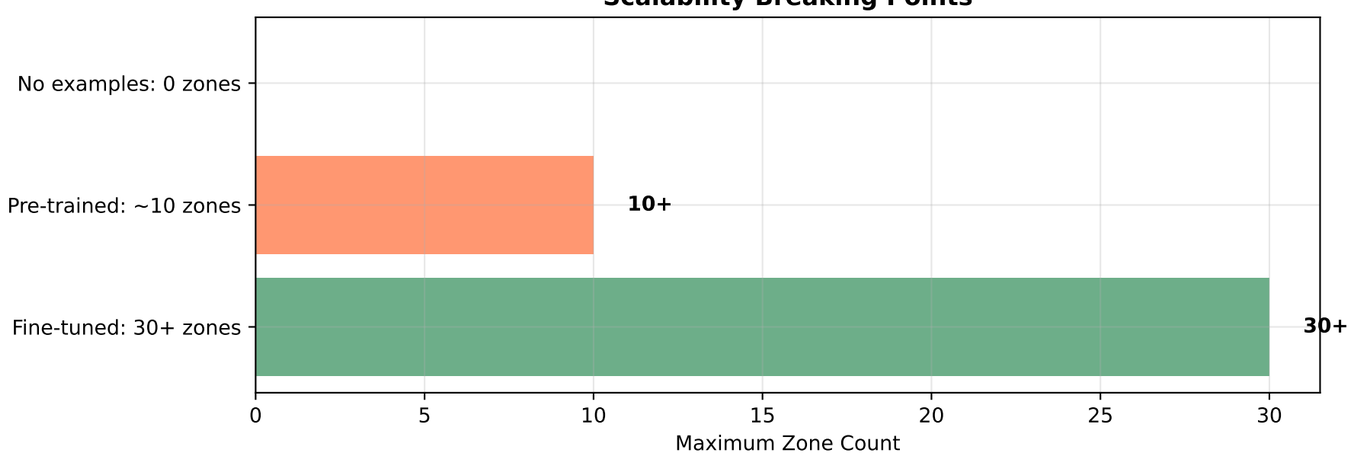
First-try Success (Fine-tuned): 75%

Avg Iterations (Fine-tuned): 0.2

Output Quality Matrix (Zone Count vs Model)



Scalability Breaking Points



Key Research Findings

- ✓ **Fine-tuning provides 5x performance improvement**
- ✓ **100% success rate across all building scales (1-30 zones)**
- ✓ **Zero post-processing required (clean IDF output)**
- ✓ **Consistent performance without quality degradation**
- ✓ **Domain-specific training essential for engineering tasks**